SOFTWARE TESTING REPORT

TUTORIAL – 11: PYTHON AUTOMATION TESTING

Project Detail	Information
Student Name	Parth Fegade
Enrollment Number	2504030100083
Programme	BCA — Semester 3
Practical	Tutorial – 11: Python Automation Testing
Testing Tools	Python, Selenium WebDriver, PyCharm, Google Chrome
Application Under Test	SauceDemo — saucedemo.com
Date	29 October 2026

1. Aim

To analyze and perform automation testing in Python for website navigation, login forms, links, product selection, cart navigation, checkout forms, multi-page workflows, validation, and regression-testing scenarios.

2. Objectives

- Open the SauceDemo website in Google Chrome using Selenium WebDriver.
- Locate the username and password fields and submit valid login credentials.
- Verify that login navigates to the inventory/products page.
- Select a product and verify cart navigation.
- Complete the checkout form and validate the next checkout page.
- Test browser back and forward navigation.
- Display clear test outcomes in the PyCharm console and retain screenshots as evidence.

3. Tools and Technologies

Tool / Technology	Purpose
Python	Programming language used to write the automation script.
Selenium WebDriver	Automates browser actions and validates web page behavior.
PyCharm IDE	Environment used to create and execute the Python script.
Google Chrome	Browser used for the test execution.
WebDriverWait / Expected Conditions	Waits for elements or URL conditions before continuing.
SauceDemo	Demo e-commerce website used as the application under test.

4. Implementation Details

Parameter	Details
Project Name	Python Automation Testing
Application / Software	SauceDemo Web Application
Module	Navigation, Forms, Links and Multi-Page Workflow
SDLC Model	Agile (as stated in the practical handout)
Programming Language	Python
IDE / Tool	PyCharm
Automation Tool	Selenium WebDriver
Browser	Google Chrome
Test Data	Username, password, product selection and checkout details

5. Test Case Details

The test cases below follow the scenarios and expected outcomes described in Tutorial 11.

Test Case ID	Scenario	Expected Result	Observed Result	Status
TC-11-01	Open website	SauceDemo opens in Chrome	Website opened successfully (console)	Pass
TC-11-02	Valid login	User reaches inventory page	Login test reported PASS	Pass
TC-11-03	Product selection	Selected product is added to cart	Product selection reported PASS	Pass
TC-11-04	Open cart	Cart page opens	Cart navigation reported PASS	Pass
TC-11-05	Checkout form	Checkout data is accepted	Checkout form reported PASS	Pass

TC-11-06	Multi-page validation	Correct checkout page opens	Multi-page workflow reported PASS	Pass
TC-11-07	Back/forward navigation	Browser navigates back and forward	Back/forward navigation reported PASS	Pass

6. Test Procedure

1. Open PyCharm and create or open the Python project.
2. Create the Python file named tutorial11.py.
3. Install Selenium in the PyCharm terminal using: pip install selenium.
4. Import webdriver, By, WebDriverWait, and expected_conditions as EC.
5. Launch Chrome with webdriver.Chrome() and maximize the window.
6. Open https://www.saucedemo.com/.
7. Enter the valid username standard_user and password secret_sauce, then click Login.
8. Wait for the URL to contain “inventory” and assert that the inventory page is open.
9. Click the Add to cart button for Sauce Labs Backpack.
10. Open the shopping cart and verify that the cart page is displayed.
11. Click Checkout and fill the first name, last name, and postal-code fields with the test data specified in the practical.
12. Click Continue and verify that the URL contains “checkout-step-two”.
13. Navigate back and forward, then close the browser with driver.quit() in a finally block.
14. Review the console output and capture screenshots of the code, execution results, login page, and product page.

7. Test Execution and Result Analysis

The supplied PyCharm console screenshot reports successful completion of the website, login, product selection, cart navigation, checkout form, multi-page workflow, and back/forward navigation checks. It also displays “Tutorial 11 Completed Successfully” and exit code 0. The supplied browser screenshots show the SauceDemo login screen with credentials entered and the inventory/products page after login.

The evidence included with this report supports the successful execution messages shown in the console. The supplied screenshots do not individually show every intermediate cart and checkout page, so those steps are recorded as reported by the console rather than independently verified from a separate browser screenshot.

8. SDLC–STLC Mapping

SDLC Phase	Corresponding STLC Phase	Testing Activity
Requirement Analysis	Requirement Analysis	Identify automation requirements.
System Design	Test Planning	Prepare automation scenarios.
Development	Test Case Development / Implementation	Develop Selenium scripts.
Testing	Test Execution	Execute automated scripts and record results.
Deployment	Test Closure	Verify and summarize test results.
Maintenance	Regression Testing	Re-run existing scenarios after changes.

9. Testing Activities and Deliverables

Activity	Deliverable / Artifact	Purpose
Identify automation requirements	Requirement List / SRS	Defines what should be tested.
Prepare automation scenarios	Test Plan	Defines the testing approach.

Develop Python Selenium script	Source Code	Automates the test cases.
Execute automated tests	Test Execution Report	Records observed test results.
Summarize testing	Test Summary Report	Communicates overall outcomes.
Re-execute important scenarios	Regression Report	Checks existing functionality after changes.

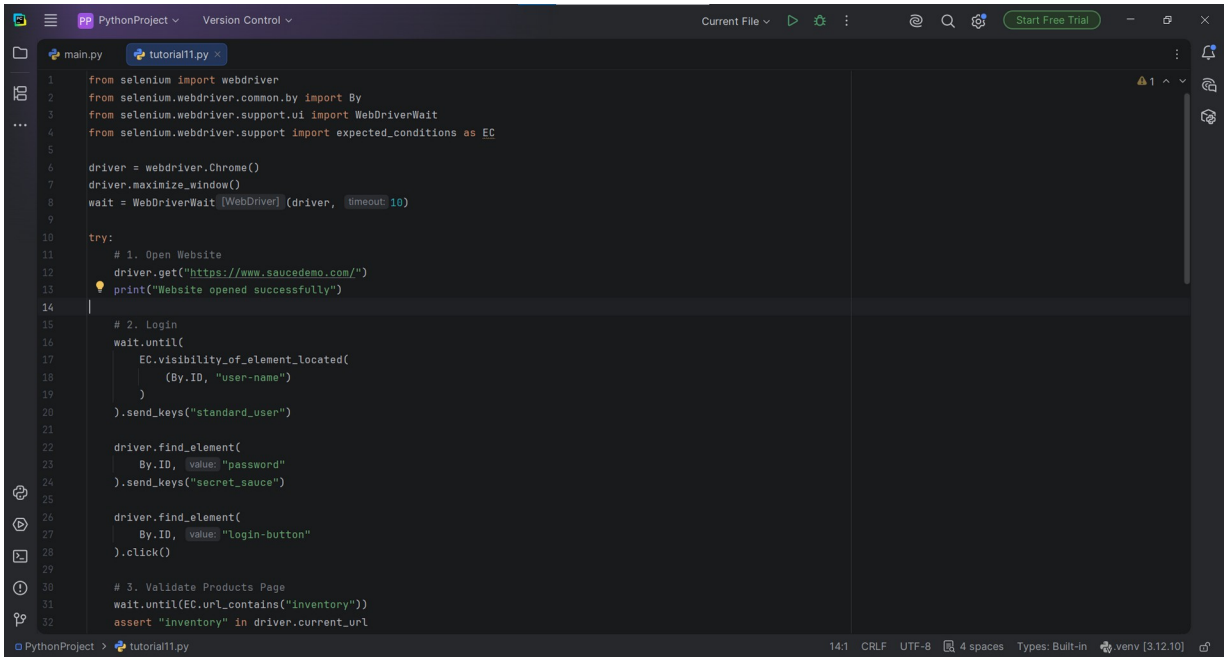
10. Roles and Responsibilities

Role	Responsibility	Phase Involved
Project Manager	Manages project activities	All phases
Business Analyst	Defines requirements	Requirement
System Analyst	Analyzes the system	Requirement / Design
Software Developer	Develops software	Development
Software Tester / QA Engineer	Performs testing	Testing
Database Administrator	Manages database	Development / Testing
Client / End User	Provides feedback	Testing / Deployment

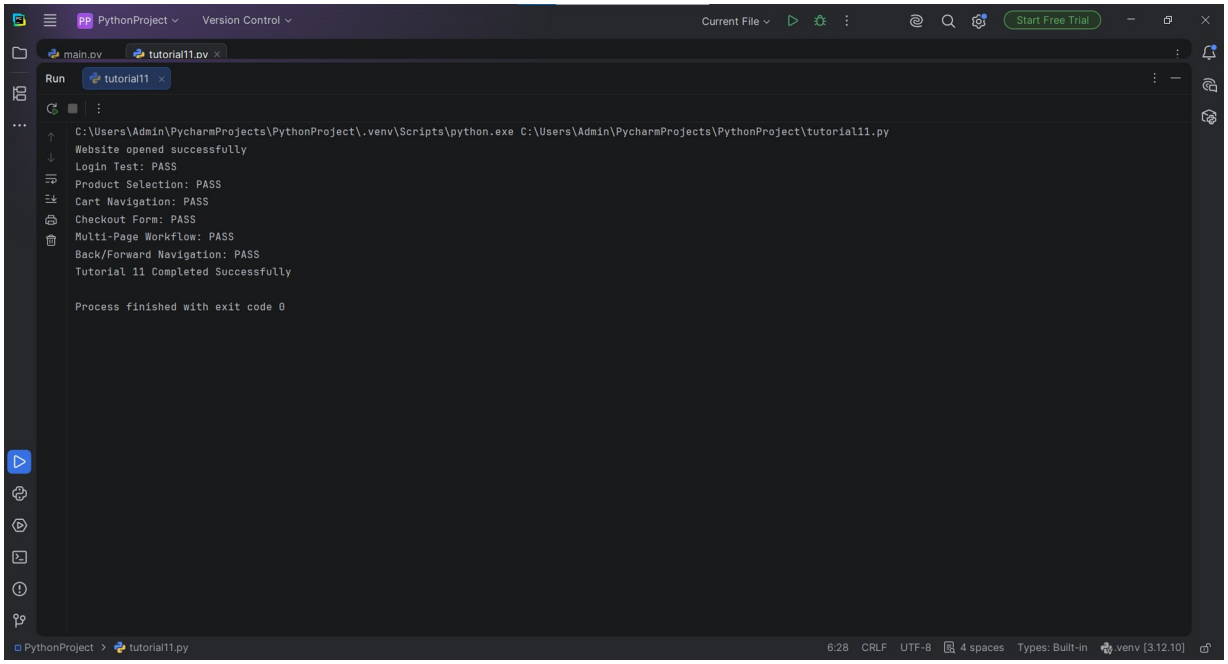
11. Screenshot Evidence

The following images were supplied for this report and are included as execution evidence.

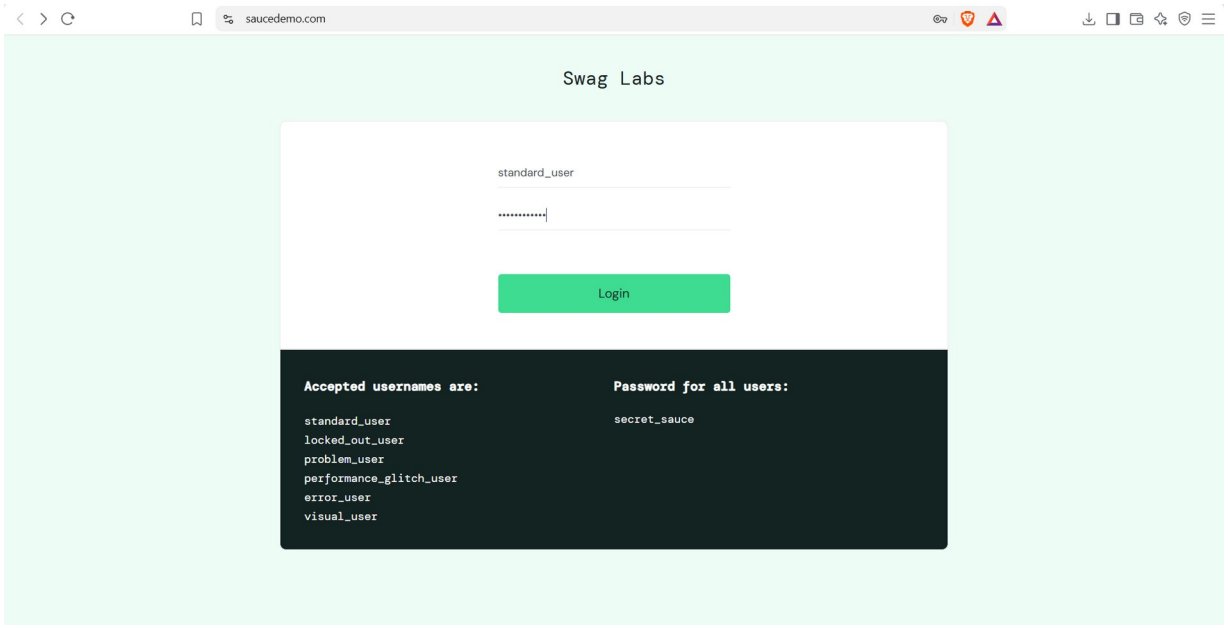
Screenshot 1 — PyCharm source code for Tutorial 11.



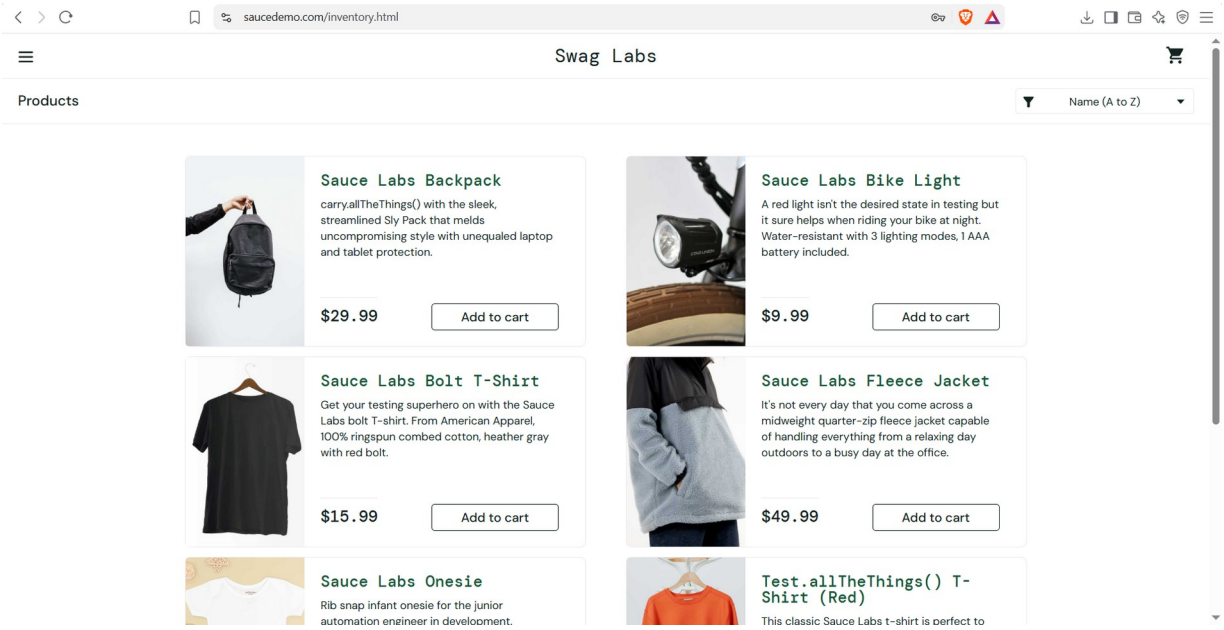
Screenshot 2 — PyCharm console output showing the test results.



Screenshot 3 — SauceDemo login page with the test credentials entered.



Screenshot 4 — SauceDemo inventory/products page after login.



12. Observation

Sr. No.	Observation	Remarks
1	Chrome was launched through Selenium.	Successful — console output
2	Login form was automated.	Successful — console output
3	Product selection was executed.	Reported successful — console output
4	Cart navigation was executed.	Reported successful — console output
5	Checkout form was completed.	Reported successful — console output
6	Multi-page workflow was validated.	Reported successful — console output
7	Back/forward navigation was executed.	Reported successful — console output

13. Conclusion

Tutorial 11 demonstrates Python-based web automation testing with Selenium WebDriver on the SauceDemo application. According to the supplied console output, the website navigation, login, product selection, cart navigation, checkout form, multi-page workflow, and back/forward navigation checks all passed. The screenshots provide supporting evidence of the Python script, console results, login form, and inventory page.